Guidelines to formulate OTC drugs in topical emulsions with improved sensorial properties

E. DUBOIS, E. DAUPHIN-CHANARD, D. HOSTE, D. MARCHAUD

GATTEFOSSÉ - 36 chemin de Genas - BP 603 - F-69804 Saint Priest Cedex - FRANCE

PURPOSE

The main objective of a dermal formulation is to carry the active pharmaceutical ingredient (API) across the skin barrier and to the site of action. But it should also display satisfying sensorial properties to drive patient compliance and product differentiation.

It is therefore important to provide formulators with guidelines to facilitate the selection of the right emulsifier, ie. one which provides excellent emulsification and drug delivery properties in addition to having excellent textural properties and delivering a pleasurable sensorial aspect when applied on the skin.

INGREDIENTS

Gattefossé excipients

Trade name	Chemical name	Functionality
Apifil® **Gelot™ 64** **Sedefos™ 75** **Tefose® 63**	*PEG-8 beeswax* *Glyceryl stearate (and) PEG-75 Stearate* *Triceteareth-4 phosphate (and) glycol stearate (and) PEG-2-stearate* *PEG-6 stearate (and) glycol stearate (and) PEG-32 stearate*	*O/W emulsifier*
Plurol® Diisostearique	*Polyglyceryl-3-diisostearate*	*W/O emulsifier*
Emulcire 61 WL 2659 **Plurol® Oleique CC497**	*Cetyl alcohol (and) ceteth-20 (and) steareth-20* *Polyglyceryl-3 dioleate*	*Co-emulsifier*
Labrafil® M1944CS	*Oleoyl Polyoxylglycerides*	*Oily phase*

Other ingredients

Excipients used as oily phase were mineral oil (supplied by Aiglon S.A.) and sweet almond oil (Cooper). Glycerine (Peter Cremer GmbH) was used as hydrophilic solvent and cetyl alcohol (Cognis) as texture agent. Carbomer (Lubrizol) was selected as gelling agent, sodium hydroxide (Merck) as its neutralizing agent. The preservatives used are methylparaben sodium salt (Quarrechim), methylparaben (Univar), propylparaben (Nipa) and sorbic acid (Chimidis). Stabilizers used are sodium chloride (Univar) and magnesium sulfate (Merck).

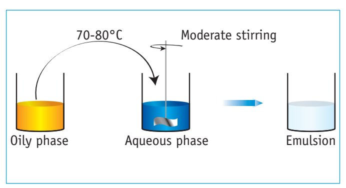
METHODS

Emulsification process

Emulsions are made by either a classical process or by a 'one-pot' process.

The classical process:

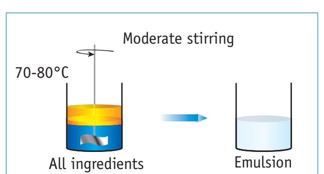
- Hot process: the aqueous and oily phases are weighed and separately heated (70-80°C).
  The aqueous phase is poured into the oily phase under moderate stirring (350 rpm). Following emulsification, cool down to room temperature using a cold water bath under stirring.



- Cold process: the aqueous and oily phases are weighed and combined under faster stirring (3000 rpm) at room temperature.

The one-pot process:

- The aqueous and oily phases are weighed and combined in the same container and heated (70-80°C), under gentle stirring. Following emulsification, cool down to room temperature using a cold water bath under stirring.



Characterization of the emulsion

An emulsion should be characterized 24hrs after production to allow sufficient maturation time.

Macroscopic evaluation
To evaluate the aspect, the colour and the consistency of the emulsion.

Microscopy
To assess the microstructure of the emulsion under a stereomicroscope. Determine whether the emulsion is homogeneous or heterogeneous, and fine or coarse. Observation under polarized light enables detection of crystals.

pH
This evaluation is performed using a pHmeter only if the emulsion displays an aqueous external phase.

Rheological evaluation
To measure the viscosity of the emulsion (using a viscometer Tve-05 at 200 rpm at 25°C), which is associated with its microstructure (ie. the size and the shape of the globules) and its physicochemical state.

Sensorial profile
A sensorial panel composed of trained experts evaluates the specific sensorial properties of each emulsion by a scoring technique:

- Immediate properties (ie. thickness, consistency)
- Rub-in properties (ie. greasiness, softness, spreadability)
- After-feel properties (ie. film-residue, greasy/soft/gloss afterfeel).

The results are plotted on to a spider diagram resulting in a sensorial map.

Stability study
Stability tests are performed according to ICH conditions.

A physical chemical change within formula is indicative of a stability issue.

GATTEFOSSÉ 'DERMACARE' PLACEBO FORMULATIONS

Gattefossé has 5 emulsifiers with excellent properties for pharmaceutical topical formulations.

We have developed a 'Dermacare' kit to demonstrate the physico-chemical and the sensorial properties of these emulsifiers.

This kit comprises 7 formulation samples, each containing one of the following emulsifiers: Apifil®, Gelot™ 64, Plurol® Diisostearique, Sedefos™ 75 and Tefose® 63. Each formulation has been evaluated and selected for its enhanced sensorial properties (determined by sensorial analysis and mapping).

Five of them are presented hereafter as well as their characteristics.

	OilPhase◆Plus	Challenge◆Plus	WinOCold◆Plus	Anhydro◆Plus	Microbicide◆Plus
Process	Classical (hot)	Classical (hot)	Classical (cold)	One pot	Classical (hot)
Macroscopic	Compact white cream	Semi-compact white cream	Compact and bright white cream	Very compact and slightly yellow translucent emulsion	Semi-fluid white cream
Microscopic	2-10 µm	6-12 µm	Homogeneous Fine	Homogeneous	1-3 µm
pH	5.3	4.5	/	/	4.7
Viscosity (mPa.s)	5110	3910	4225	9360	2640
On-going stability @ (40°C)	6 months	6 months	6 months	3 months	6 months

OilPhase◆Plus

Ingredients	Functionality	%W/W
Apifil®	**Emulsifier**	**5.00**
Emulcire™ 61 WL 2659	**Co-emulsifier**	**3.00**
Mineral oil	Oily phase	30.00
Demineralized water	Aqueous phase	61.60
Carbomer	Gelling agent	0.10
Sodium hydroxide (10% sol.)	Neutralizing agent	0.20
Sorbic acid	Preservative	0.05
Methyl paraben sodium salt	Preservative	0.05

Apifil® is a powerful emulsifier ideal for formulations incorporating a high volume of oil and lipophilic API.

Challenge◆Plus

Ingredients	Functionality	%W/W
Gelot™ 64	**Emulsifier**	**3.00**
Emulcire™ 61 WL 2659	**Co-emulsifier**	**3.00**
Cetyl alcohol	Texture agent	3.00
Labrafil® M1944CS	**Oily phase**	**2.00**
Mineral oil	Oily phase	10.00
Demineralized water	Aqueous phase	78.85
Sorbic acid	Preservative	0.10
Methyl paraben sodium salt	Preservative	0.05

Gelot™ 64 is a trouble-shooting emulsifier for difficult to formulate API including acids and alcohol extracts.

WinOCold◆Plus

Ingredients	Functionality	%W/W
Plurol® Diisostearique	**Emulsifier**	**3.00**
Plurol® Oleique CC497	**Co-emulsifier**	**2.00**
Mineral oil	Oily phase	15.00
Methyl paraben	Preservative	0.25
Propyl paraben	Preservative	0.15
Demineralized water	Aqueous phase	77.60
NaCl	Stabilizer	1.00
$MgSO_4 .7H_2O$	Stabilizer	1.00

Plurol® Diisostearique is a PEG-free, W/O emulsifier ideal for heat-sensitive API when used with a cold process (a hot process formulation is also feasible: ask for WinOHot•Plus).

Anhydro◆Plus

Ingredients	Functionality	%W/W
Sedefos™ 75	**Emulsifier**	**9.00**
Sweet almond oil	Oily phase	25.00
Glycerine	Aqueous phase	66.00

Sedefos™ 75 is a problem-solving emulsifier for highly hydrosensitive API. It also could be used in regular emulsion for difficult-to-formulate active (ask for Resolv•Plus).

Microbicide◆Plus

Ingredients	Functionality	%W/W
Tefose® 63	**Emulsifier**	**8.00**
Mineral oil	Oily phase	18.00
Cetyl alcohol	Texture agent	2.00
Demineralized water	Aqueous phase	71.90
Sorbic acid	Preservative	0.05
Methyl paraben sodium salt	Preservative	0.05

Tefose® 63 is a leading emulsifier for anti-fungal treatment due to its excellent mucosal tolerance.

GUIDELINES TO FORMULATE OTC DRUGS

The initial approach for selecting the right emulsifier for the drug is by drug solubility screening. Two questions have then to be answered:

- How much solubilizer (either hydrophilic or lipophilic) is required to solubilize the drug?
- Which emulsifier can emulsify that amount of solubilizer?

Gattefossé has determined the maximum amount of lipophilic or hydrophilic phases that each emulsifier is able to disperse. This evaluation was done for emulsifier concentration corresponding to those of the Dermacare placebo formulations. The results are presented in the following tables.

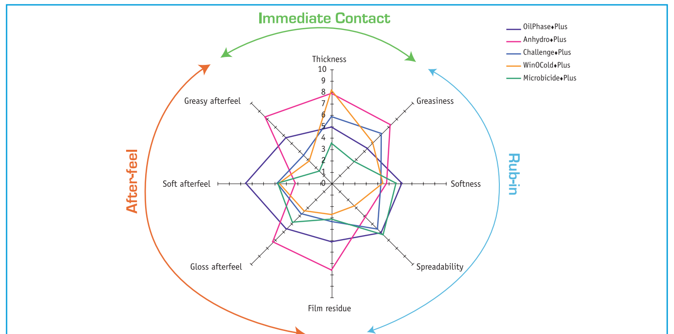
Matching emulsifiers with lipophilic phases

Max amount of excipient to be emulsified by:	Medium Chain Triglyceride	Mineral oil	Sweet Almond oil	Labrafil®
Apifil®	*30%*	*40%*	*30%*	*30%*
Gelot™ 64	*12%*	*30%*	*20%*	*12%*
Tefose® 63	*18%*	*18%*	*18%*	*10%*
Sedefos™ 75	*25%*	*40%*	*25%*	*25%*
Plurol® Diisostearique	*< 4 %*	*30 %*	*< 4 %*	*< 4 %*

Matching emulsifiers with hydrophilic phases

Max amount of excipient to be emulsified by:	Glycerine	Propylene glycol	Ethanol	Transcutol® P	PEG 300
Apifil®	*15%*	*20%*	*10%*	*< 10%*	*< 10%*
Gelot™ 64	*20%*	*20%*	*10%*	*10%*	*< 5%*
Tefose® 63	*5%*	*< 30%*	*10%*	*10%*	*5%*
Sedefos™ 75	*Complete substitution*	*20%*	*10%*	*< 20%*	*< 20%*
Plurol® Diisostearique	*25%*	*< 3%*	*5%*	*10%*	*< 5%*

Another way of selecting the right emulsifier can be driven by the sensorial identity of the emulsion it can be built. The sensorial properties of the placebo Dermacare formulations are reported in the spider diagram below.



Understanding the sensorial properties of emulsifiers now enables the selection of emulsifiers as well as the optimization of formulations based on the 'sensorial' objectives of the formulation, for example for the treatment of irritated, damaged skin conditions requiring frequent applications a topical formulation that has good spreadability and softness is clearly preferable.